Abu Dhabi City Campus Daily/Weekly Planner

Week 11: 9th – 13th November 2026

Subject: Mathematics Diploma Programme **Unit Title: Functions and Complex Numbers** **Grade: 11**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Domain and Range	I can determine the domain and range of a function from its equation, graph and context, using appropriate restrictions and interval notation.	2.1 Clarify domain, range, input, output, interval notation and restriction using labelled examples. 2.3 Provide a step-by-step decision guide and multiple representations (mapping, graph, equation and context).	Relevant Videos Functions, domain, range [IB Maths AI SL/HL] Inverse functions [IB Maths AI SL/HL]
Lesson 2: Functions and Inverse Functions	I can determine whether a relation has an inverse function and find and verify the inverse of a one-to-one function algebraically and graphically.	2.3 Provide an inverse-finding checklist and colour-code the interchange of x and y. 5.3 Scaffold with linear examples before progressing to quadratic or rational functions.	
Lesson 3: Applications of Functions and Inverses	I can use functions and inverse functions to model and solve contextual problems, interpret the meaning of inputs and outputs, and explain the limitations of a model.	5.3 Give structured contexts and partially completed models for support. 8.2 Extension: compare two possible models or analyse when an inverse is not meaningful.	
Lesson 4 (HL): Complex Numbers – Polar Form, Euler Form and Applications	I can represent complex numbers in Cartesian, polar and Euler forms, convert between these forms, and apply modulus and argument to solve mathematical problems involving complex numbers.	8.2 Vary the level of challenge through differentiated tasks: 9.3 Develop self-assessment and reflection through peer checking, worked-solution comparisons and an exit ticket in which students identify one strength and one area for improvement.	
Homework	Complete homework assigned on InThinking. SL: Domain, Range, Functions and Inverses. HL: Complex Numbers.		
Teacher's notes / Reminders	Students must bring a device to access digital resources, a graphing calculator and a notebook.		

Technology Integration	Microsoft Teams/OneNote for lesson notes and assignments; GDC/Desmos for graphing functions, inverse reflections and Argand diagrams
Cross-curricular links	**Physics/Engineering:** complex numbers in alternating-current analysis and wave representation. **Computing:** functions as input-output processes and reversible operations.

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External Assessment Integration	International Baccalaureate Mathematics Applications & Interpretation SL and HL
Wellbeing/ ATL Skills	**Thinking:** analyse function properties, restrictions and inverse relationships; select strategies for complex-number conversions and powers/roots. **Communication:** use correct notation, labelled graphs and clear mathematical explanations. **Research:** identify relevant information and variables in contextual problems. **Self-management:** organise multi-step working and check answers. **Social:** collaborate and explain alternative solution methods.
Standards	**IB Mathematics Diploma Programme: Topic 2 Functions** **SL:** functions, domain and range, inverse functions and applications **HL:** complex numbers, modulus and argument, polar form, Euler form.
Inquiry/ TOK Question	How does changing the representation of a mathematical object make a problem easier to solve? Why can a function be reversible mathematically but not always meaningful in a real-world context? In what ways does Euler form reveal relationships that are less visible in Cartesian form?
Success Criteria (Measurable)	**Lesson 1 – Domain and Range** • **All students must** identify the domain and range from a mapping diagram or graph. • **Most students should** determine domain and range algebraically, applying restrictions and interval notation. • **Some students could** justify restrictions for unfamiliar functions and compare representations. **Lesson 2 – Functions and Inverse Functions** • **All students must** explain what an inverse function does and identify a one-to-one function from a graph. • **Most students should** find an inverse algebraically and verify the relationship graphically or by composition. • **Some students could** determine a suitable domain restriction and justify when an inverse is a function. **Lesson 3 – Applications of Functions and Inverses** • **All students must** identify inputs, outputs and variables in a contextual function. • **Most students should** use a function or inverse to solve a contextual problem and interpret the result with units. • **Some students could** evaluate assumptions, restrictions and limitations of the model. **Lesson 4 (HL) – Complex Numbers: Polar Form, Euler Form and Applications** • **All students must** identify the modulus and argument of a complex number and represent it on an Argand diagram. • **Most students should** convert between Cartesian, polar and Euler forms, using radians and the correct quadrant. • **Some students could** apply polar and Euler forms to multiplication and division of complex numbers and explain the geometric meaning of the result.
National Identity	Area: UAE National Identity
	Implementation
	Highlight the role of complex numbers in engineering, electrical systems, telecommunications and technological innovation relevant to the UAE's development. Encourage students to connect abstract mathematical representations to applications while evaluating the assumptions of each model.